

*2014 Metal Recycling Conference
& Achievement Conference)*

ReMat (Recycling & Materialization) for rare earths using solid-liquid reaction

**Korea Institute for Rare Metals (KIRAM)
Korea Institute of Industrial Technology (KITECH)**

Introduction

– RM & KIRAM –

DEFINITION of RARE MATERIALS

Difficult to extract from ores, but need to redefine using following 3 factors

▪ Limited natural resource
▪ rare concentration

▪ Price instability
▪ Creation of new industry

▪ Biased distribution
▪ Growth in demand

On dictionary

On economy

On security

Periodic Table of Elements

Alkali metals (6): H, Li, Na, K, Rb, Cs, Fr

Alkaline earth metals (2): Be, Mg, Ca, Sr, Ba, Ra

Transition elements (10): Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, In, Sn, Pb, Bi, Po, At, Rn

Inner transition elements (14): La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu

Rare earths (17): La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu

Refractory metals (11): Nb, Ta, Ti, Zr, Hf, Y, Sc

PGM (6): Ru, Rh, Pd, Ag, Cu, Ni, Co, Fe, Mn, Cr, V, Ti, Zr, Hf, Y, Sc

Boron group (5): B, Al, Ga, In, Tl, Sn, Pb, Bi, Po, At, Rn

Semimetals (9): Si, Ge, As, Sb, Te, Se, Te, Po, At, Rn

Other groups: He, Ne, Ar, Kr, Xe, Rn; He, Ne, Ar, Kr, Xe, Rn; He, Ne, Ar, Kr, Xe, Rn

Legend: 금속 (Metal), 준금속 (Semimetal), 비금속 (Nonmetal)

Relative concept depending on occasion of each country or era

- Korea : 54 items
- 35 species
(56 elements)
- REE: 17 elements
- PGM: 6 elements

➤ Japan : 47 items

➤ USA : 54 items

USAGE OF RARE METALS IN INDUSTRY

Car



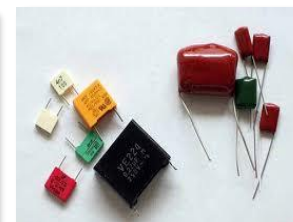
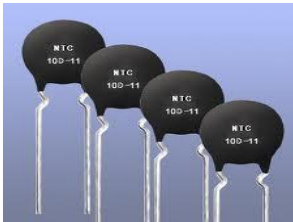
Component	Motor	Secondary battery	The steel sheet of vehicle	Exhaust pipe, Pipe	Purification of exhaust gas
Materials	REE	Ni-MH Li-ion battery	Galvanized steel sheet (GA) High speed steel (AHSS)	Stainless steel	Catalyst
Rare metals	Nd, Dy	Ni, Li, Co	Mn	Ni, Cr, Mo	Pt, Pd, Rh



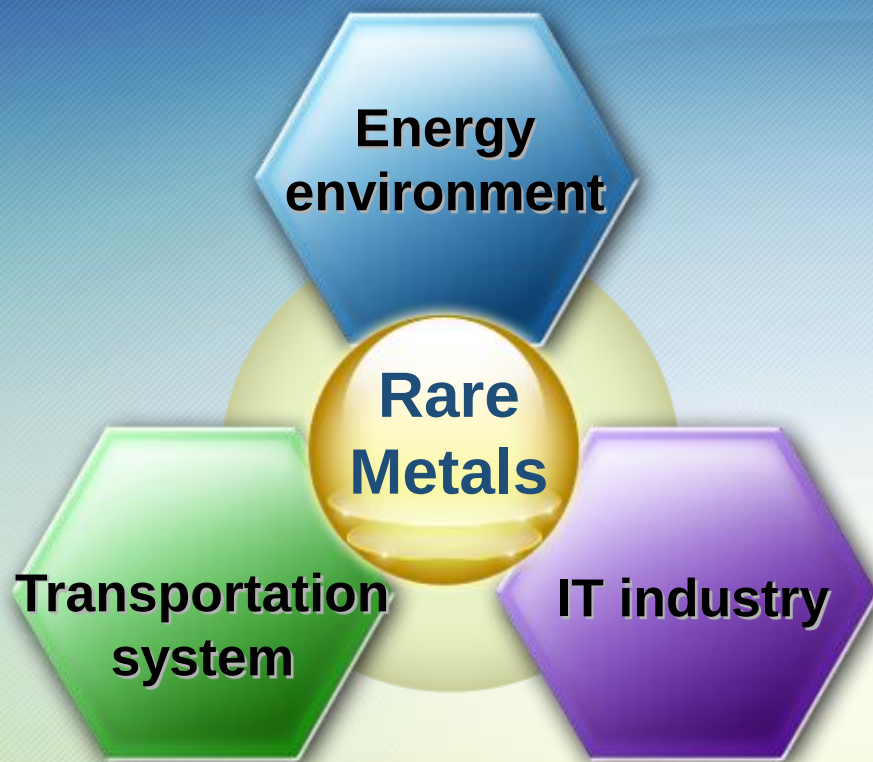
IT-Mobile



Component	Thermistor	Camera module	Li-ion, Ni-MH Fuel cell	Condenser	Small motor
Materials	Mn_3O_4 , NiO_3 , Co_3O_4	SiO_2 , InGaAs	$LiCoO_2$, $LiNiO_2$, $LiMn_2O_4$, $LiMn_2$, Li-MH	Internal electrode	Sm, Nd-based permanent magnet
Rare metals	Mn, Co, Ni	Si, In, Ga, As	Li, Ni, Co, Mn, Zr, Sb, Ba, Sr	Ta, Sn, Ni, Cu	Sm, Nd, Co



RM ON INDUSTRIES NOW & FUTURE



Energy environment

- Solar cell - In, Ga, Se, Si, Cd, Te
- Wind power - Nd, Dy, Sm, Co
- Fuel cell - Ni, Y, Zr, La, Ga, Cr

Transportation system

- Green car - REE, Li, Co, Ni
- Ship/Ocean system - Ni, Cr, W, Mo, V

IT industry

- LED - Ga, As, In, P
- Semiconductor - Si, Ga, In, P, As, Sb
- Display - Y, Eu, Tb, Ni, In

KOREA INSTITUTE FOR RARE METALS (KIRAM)

Comprehensive Driving for Rare Metals Programs

Infra structuring activities

- Establishment of total information system on
 - Science & tech. trends
 - Rare metals flow
- Construction of pilot scale R&D facilities
- Education for elementary, middle and high school students.

Enterprise supporting activities

- Assistance for R&D related issues
- Conduction of joint R&D project and industry training
- Incubation of rare metals related enterprises.
- Establishment of regional collaborating network

Global activities

- Activation of global communicating & networking
- Development of Joint R&D with foreign partners
 - MOU agreement
 - joint project conduction
 - International Standard
- Exchange of human resource and industrial strategy

TOTAL INFORMATION SYSTEM ON RARE METALS

www.kiram.re.kr



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Korea Institute for Rare Metals

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About KIRAMRare metalInformationEquipment support



Korea Institute for Rare Metals

La

Ce

Nd

Y

Pm

Sm

Eu

Co

Ni

Ga

Li

Zr

Mg

In

REE

W

Pt

Ti

Greetings

Resource-Industry-Chain of value

The Korea Institute for Rare Metals is an organization designed to establish the infrastructure for the nation's rare metals industry.



About KIRAM



Organization



Rare metal Introduction



Information



KIRAM
Korea Institute for Rare Metals

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--Affiliated Organization--

Go



산업통상자원부
MOTIE



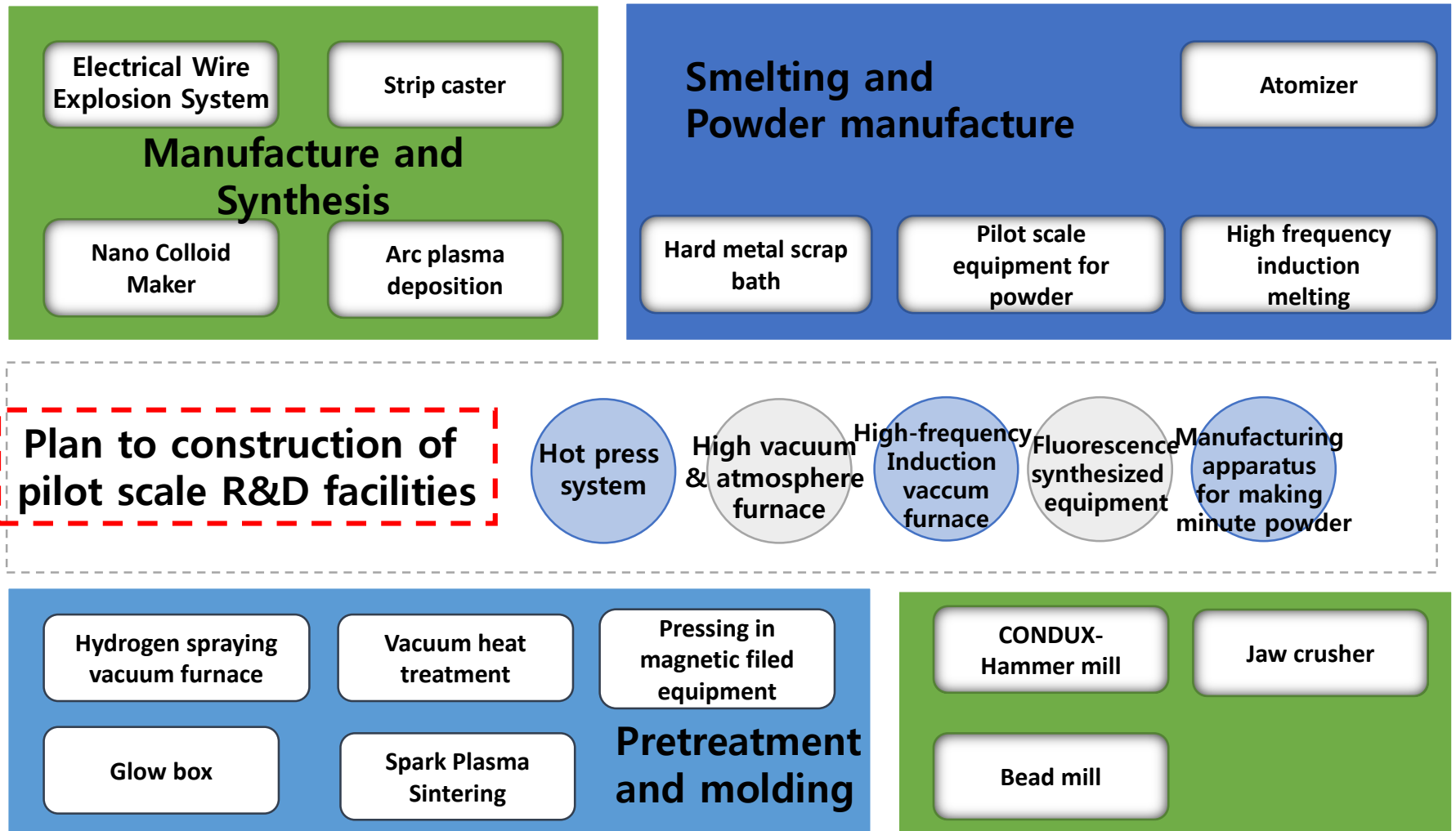
KITECH
한국형신기술연구원



KIRAM
Korea Institute for Rare Metals

RESEARCH FOUNDATION CONSTRUCTION

From process to circulation



RESEARCH FOUNDATION CONSTRUCTION

Physical· Chemical Test/Properties Analysis

Sample preparation

Centrifuge separation

polisher

Rotary evaporator

Analysis of mechanical properties

Universal testing machine

Microhardness

Linear vibrator

Analysis of structure properties

AFM

Roughness meter

Microscope

Plan to construction of pilot scale R&D facilities

AA

XRF

Portable XRF

I-V Measurement

Table SEM

VSM

Probe Station

Factsage

Analysis of physical properties

Photo luminescence

Particle size analyzer

UV-visible spectrophotometer

BET

Analysis of thermophysical properties

Laser flash analyzer

DSC

TMA

TG-DTA

Analysis of chemical properties

Micro-reactor

ICP-MS

GC-MS

R&D with RE-MATERIALIZATION (REMAT)

**Recycling and Materialization of Rare Earth
Element using Selective Extraction**

ENERGY CONSUMPTION (1) DURING SMELTING and RECYCLING

	Primary from ore (10^6 kcal/ton)	Secondary (10^6 kcal/ton)
Magnesium	90.2	3.0
Aluminum	61.5	3.0
Nickel	36.3	3.8
Copper	28.2	4.5
Zinc	16.4	4.5
Steel	8.1	3.3
Lead	6.8	3.0

*Ref. from H.S. Ray,
Central Glass & Ceramic Res. Inst .*

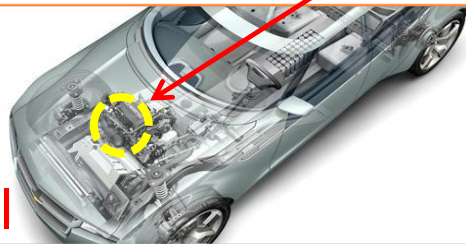
 **Recycling would be an alternative
in the aspect of resource limit and environment**

MAJOR USAGE of Nd on MAGNETS

- Industrial application of Nd-Fe-B magnet after its first use in 1982 has been rapidly expanded within past two decades..
- Creation of new applications using the Nd-Fe-B magnets

Automobile

- Motors for hybrid, electrical, fuel-battery cars
- Power windows, wiper motors
- Sensors
- Actuators



Environmental

- Electric power generator to kinetic energy (wind power, wave power)
- MHD (magneto hydro dynamic)



Medical/Bio

- Magnetic resonance imaging(MRI)
: intensity the magnetic field which determines the imaging performance



Electric/Electron

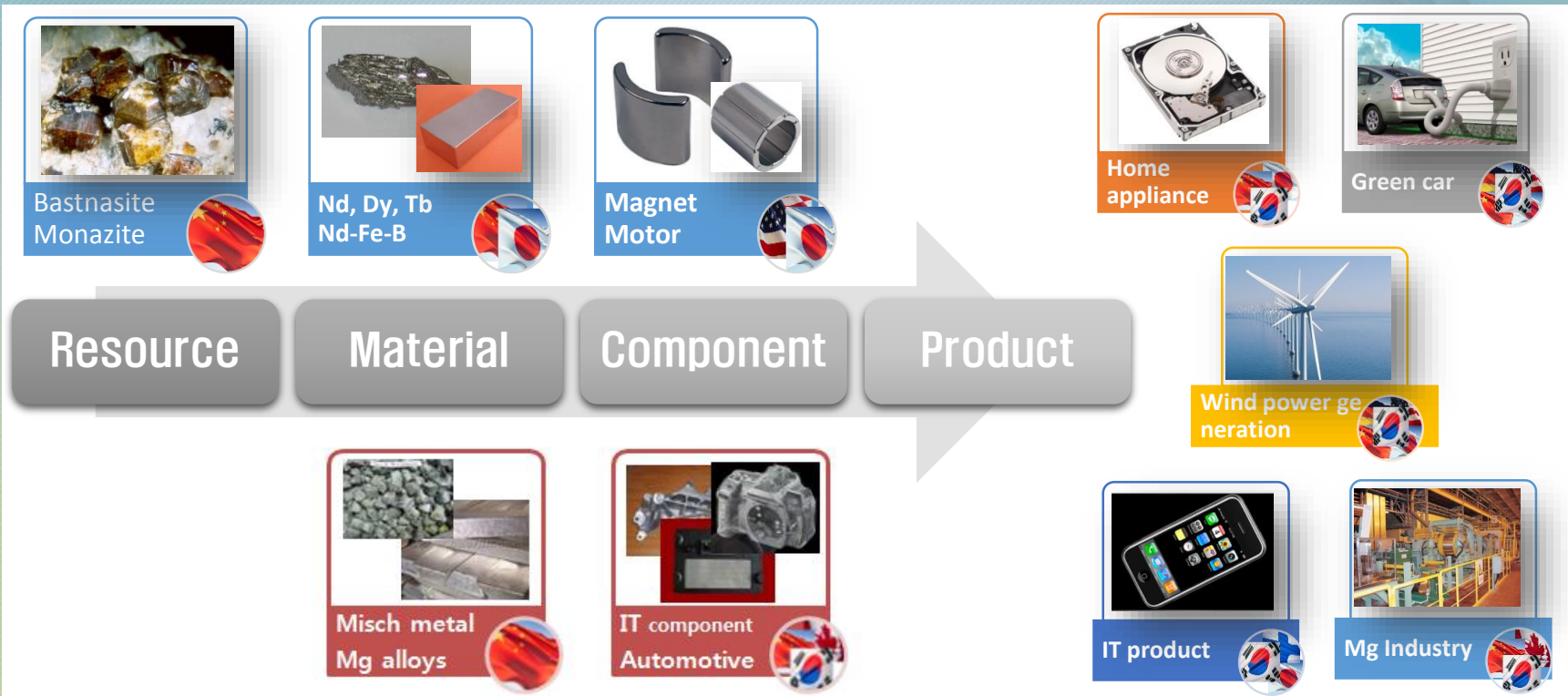
- Mobile phones & hard disc drives
- Household appliances(air conditioners, washing machines, refrigerators)
- Security systems



REE – MATERIALS FLOW FORWARD

Rare earth elements

- III B group Lanthanides (15 elements of La to Lu + Y, Sc)
- Materials hard to extract technically
- Essential materials for Green industries
such as Electric cars and Wind power generation]



CONCEPT of REE RECYCLING/MATERIALIZATION

• Recycled Materialization Technology

Emission Zero Society

Magnet to powders
circuit loop

Magnet to (Mg alloy or Nd)
circuit loop

Magnet to (Adv. or hybrid maters)
circuit loop

**Nd-Fe-B
scrap/disposal**



Mg-(Nd) alloy



**Fe-B
adv mater**

Short looped re-use

- crush-plasma treat
 - uniform RSP tech
- ⇒ Nd-Fe-B virgin powder

Selective melting - recycle

- Eco-Mg/LME process
 - Phase equilibrium & kinetics
- ⇒ Mg-Nd master alloy
⇒ Nd, Dy pure element

Fe-B remnants - recycle

- Amorphous alloy design
- Composite design

• Technology aspect

- Minimize the second contamination during recycling [**Dry-recycling tech.**]
- New environment-friendly paradigm
- Low energy/high efficiency re-use, recycle and materialization tech.

Recycling approach – LME process

Liquid Metal Extraction (LME) Processes

→ REE elements as Nd and Dy
selectively extracted to Mg melt

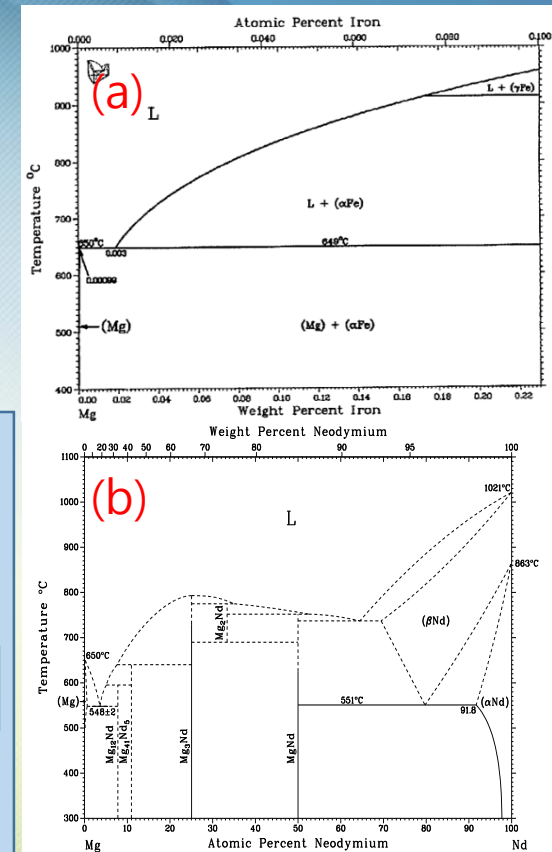
- Nd is soluble in liquid Mg and forms a number of intermetallic compounds while Fe and B are essentially immiscible in Mg.
- Liquid Mg readily takes up Nd, forming a Mg-Nd solution, while leaving the Fe and B unaffected.

Mg alloy with rare-earth elements (e.g. Nd, Y, Zr)

- improved creep resistance at elevated temp.
- reduced micro porosity during castings
- improved weld-ability

Higher cost for Mg alloying

❖ Cost effective way for Mg alloy production



The phase diagram of (a) Mg-Fe, (b) Mg-Nd

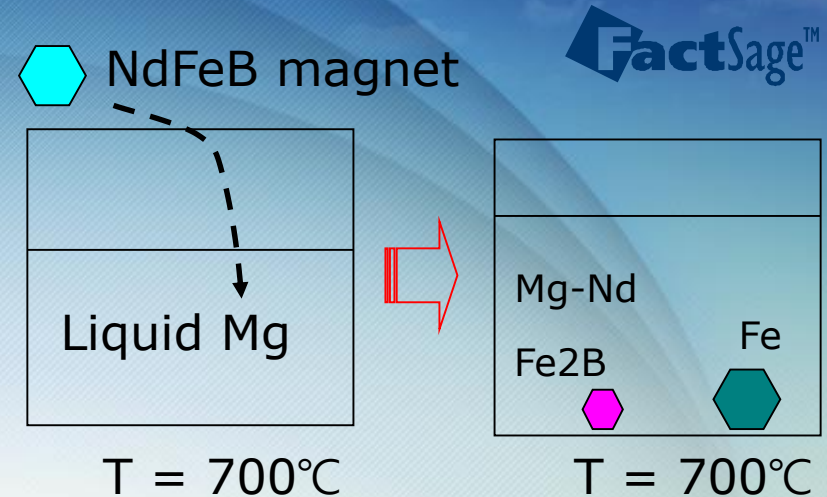
General Background & Purpose

Calculation conditions

- $T = 700\text{ }^{\circ}\text{C}$
- Nd magnet: 70wt%Nd₂Fe₁₄B + 20wt% Nd-rich + 10%Nd₂Fe₁₇

Calculation results

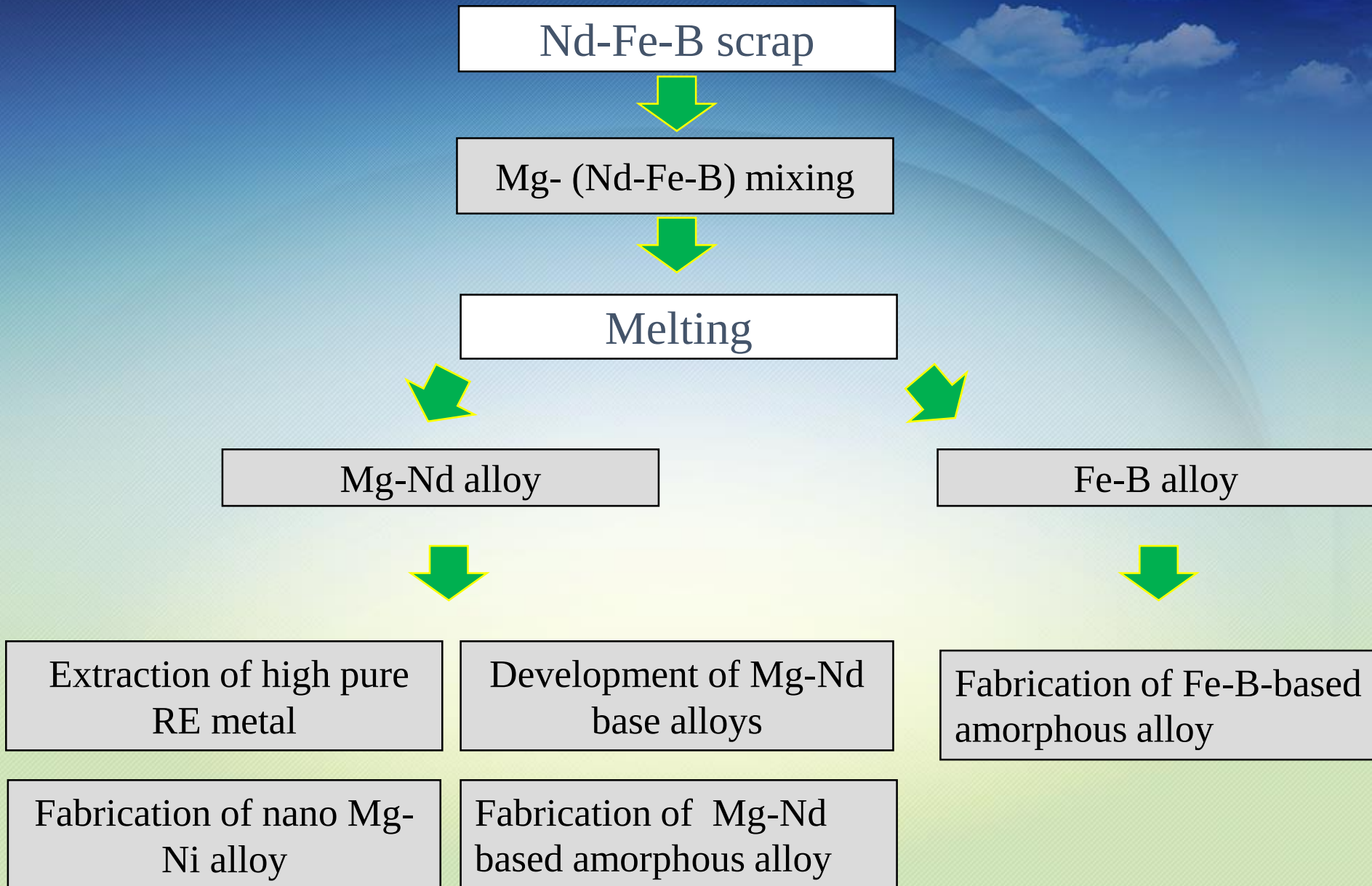
- All Nd can be extracted into liquid Mg
- Remaining phases in liquid Mg are solid Fe and Fe₂B



Is the anticipation come true?

- To identify the possibility of recycling – materialization process
- To identify the microstructure and mechanism of diffusion

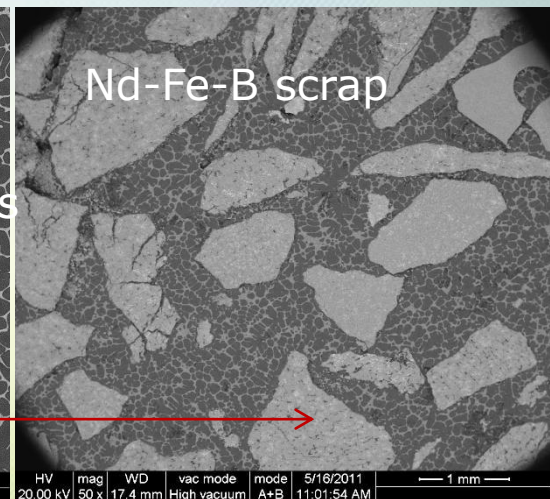
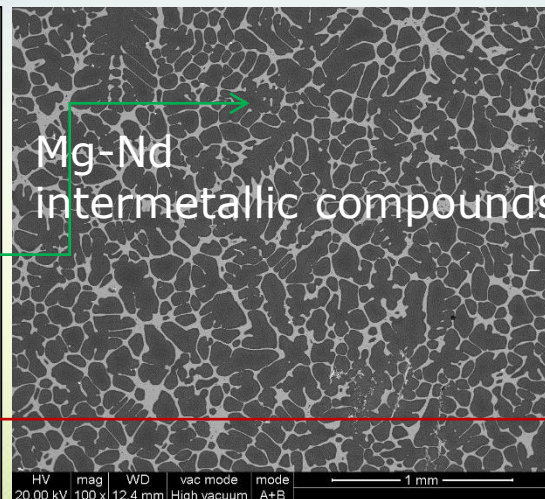
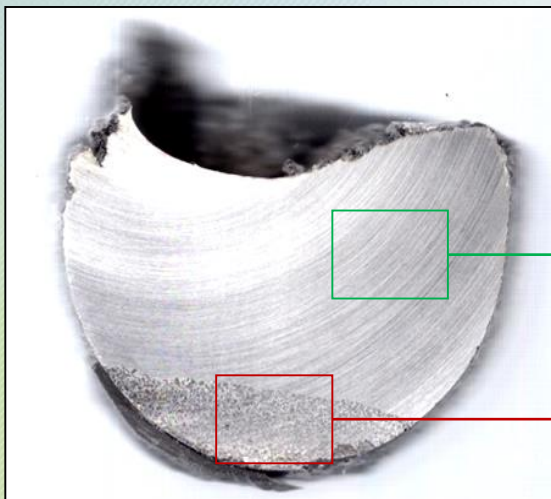
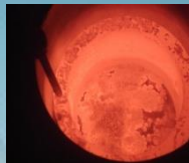
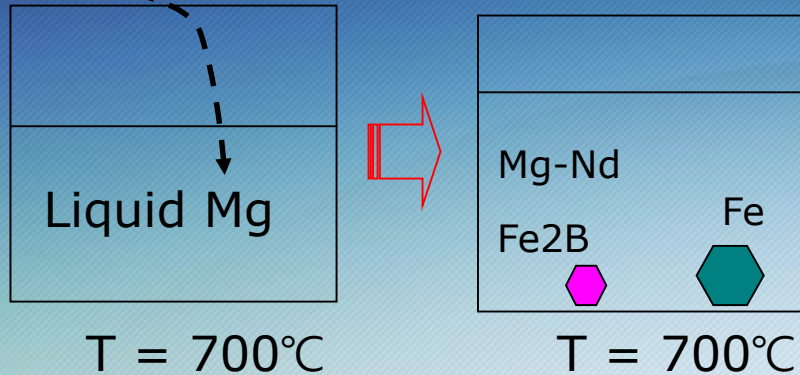
WHOLE SCHEMATIC of EXPERIMENTAL PROCEDURE



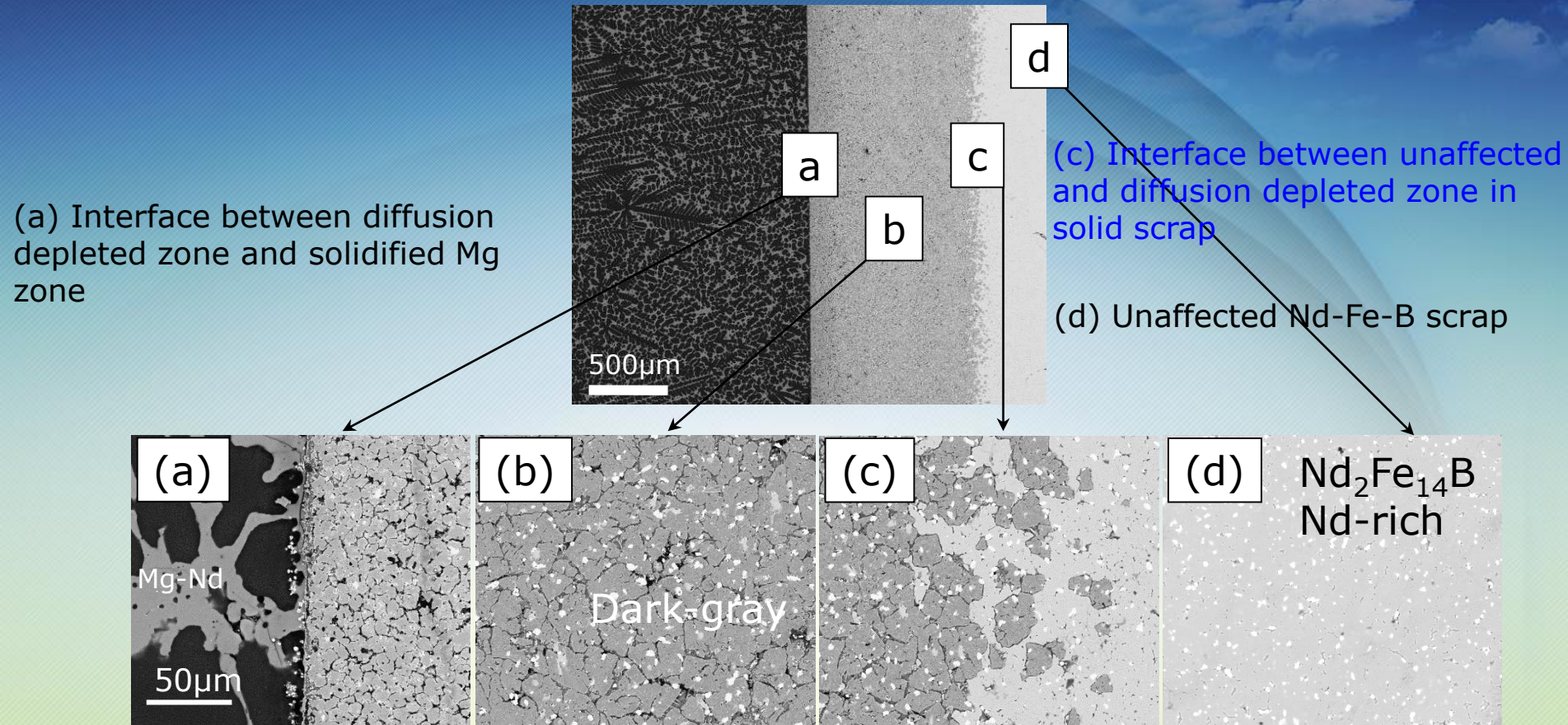
R&D CONCEPT & SAMPLES after Mg-SCRAP REACTION

FactSage™

 NdFeB magnet

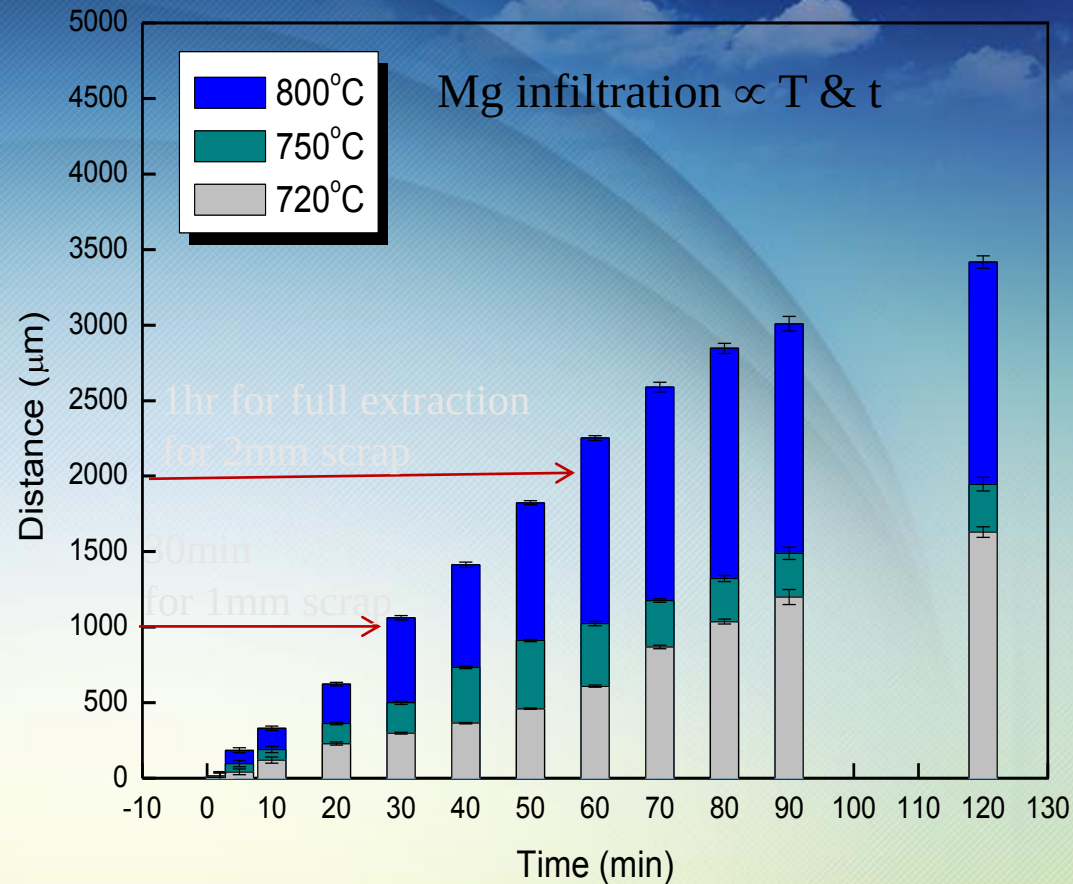
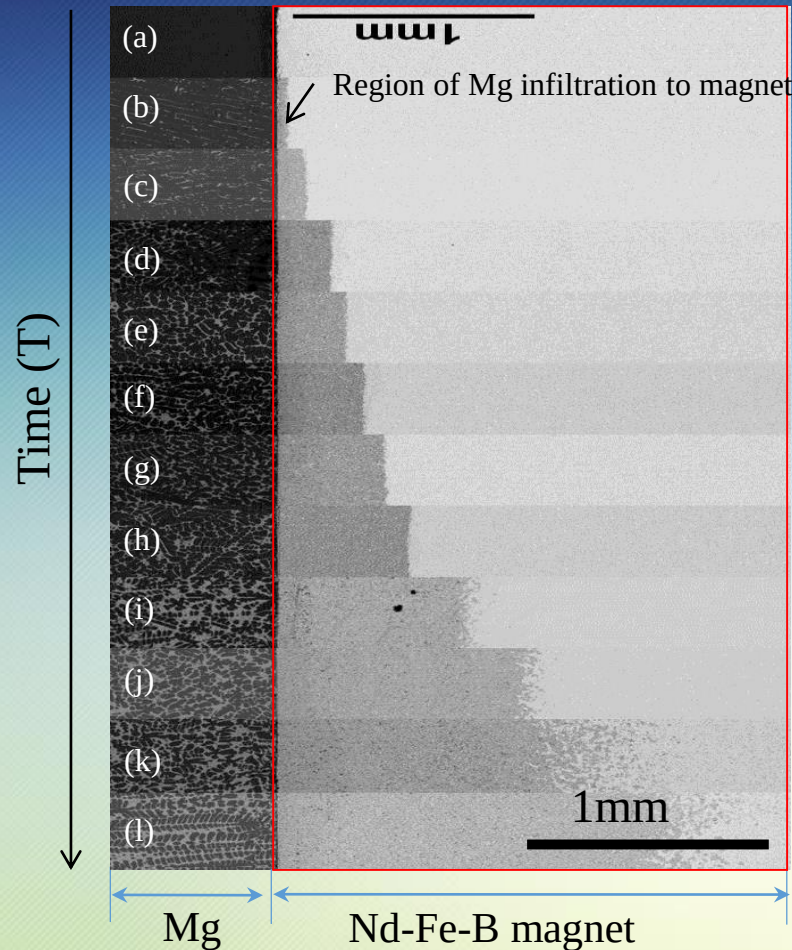


DIFFUSION of Nd to LIQUID Mg



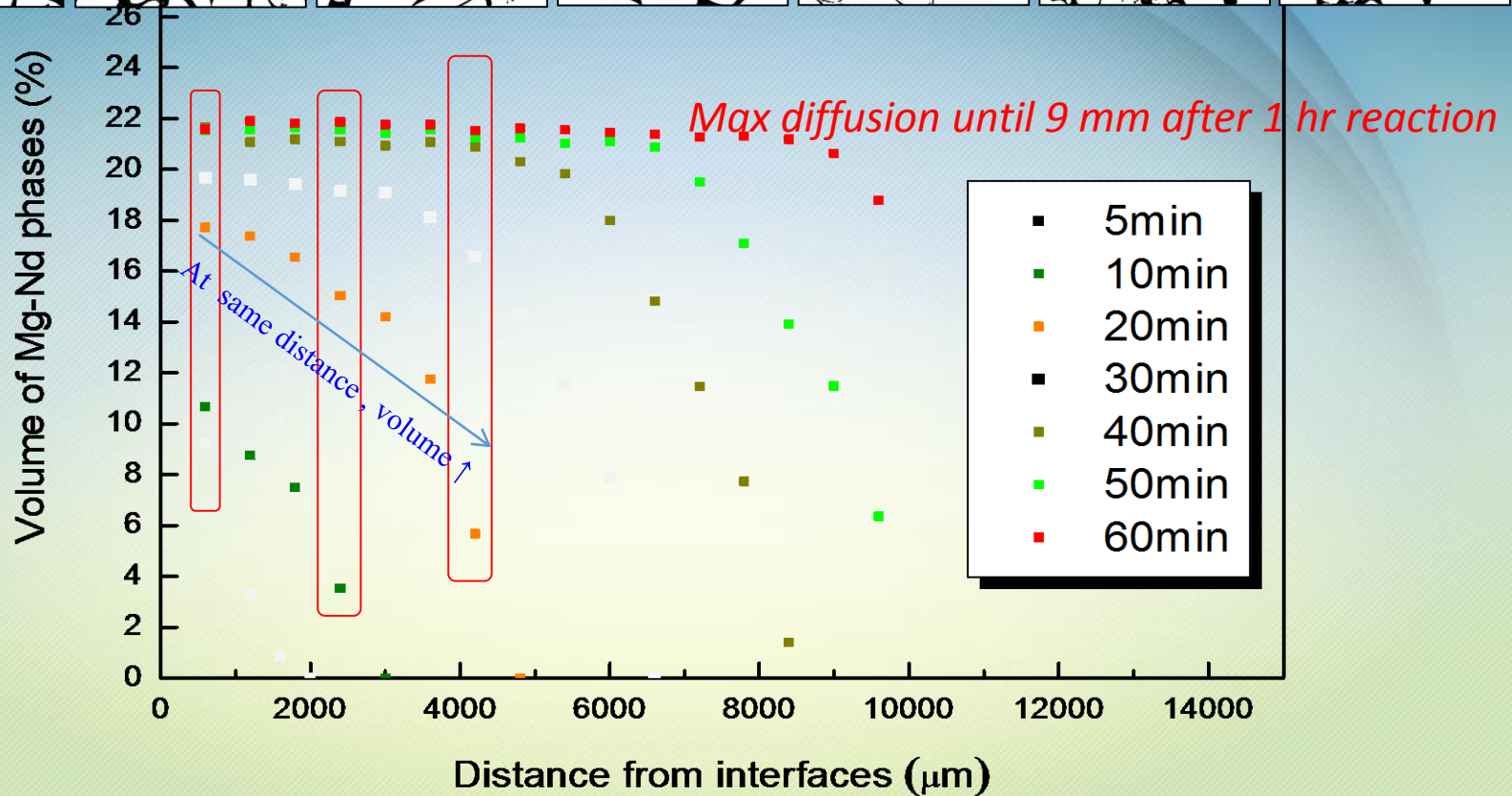
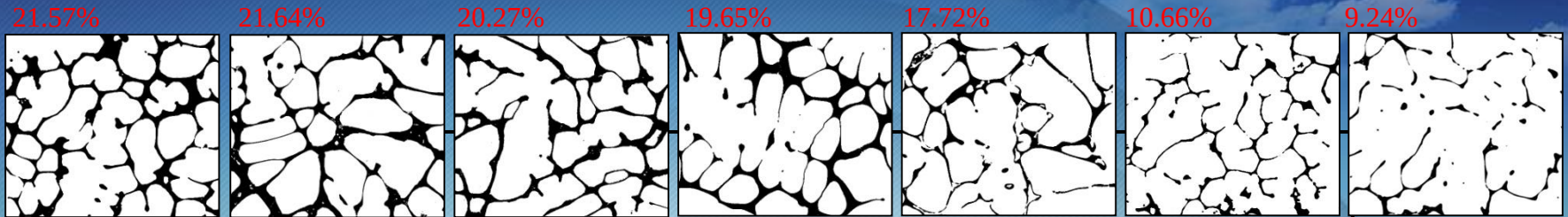
Micrographs showing the various regions from unaffected NdFeB scrap to solidified Mg

DISTANCE of Mg INFILTRATION to SCRAP with TIME & TEMPERATURE

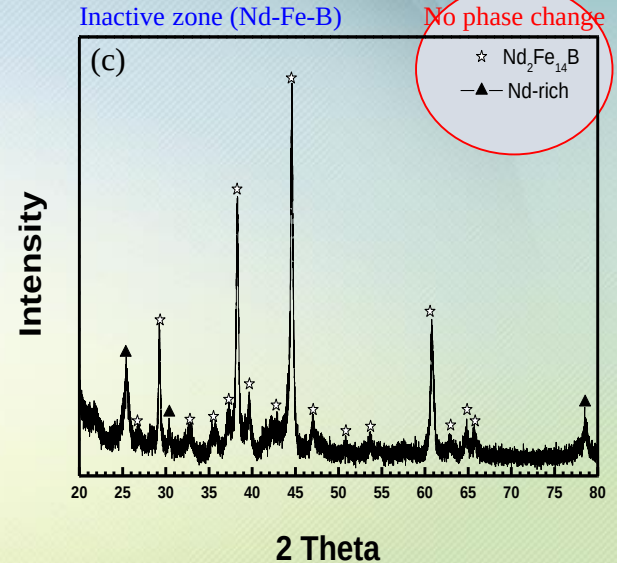
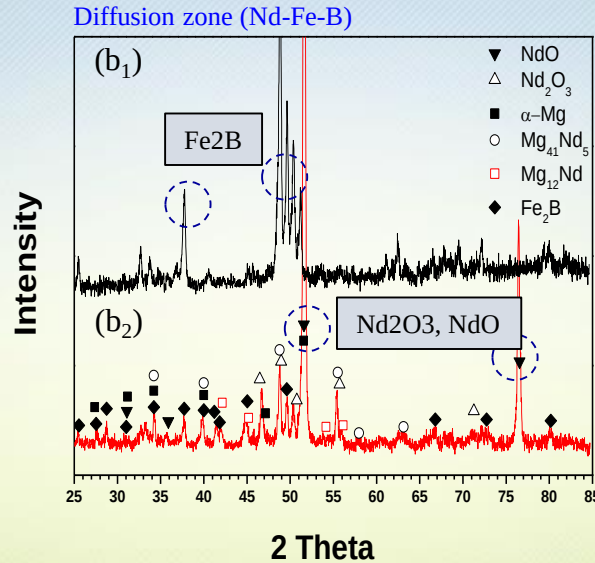
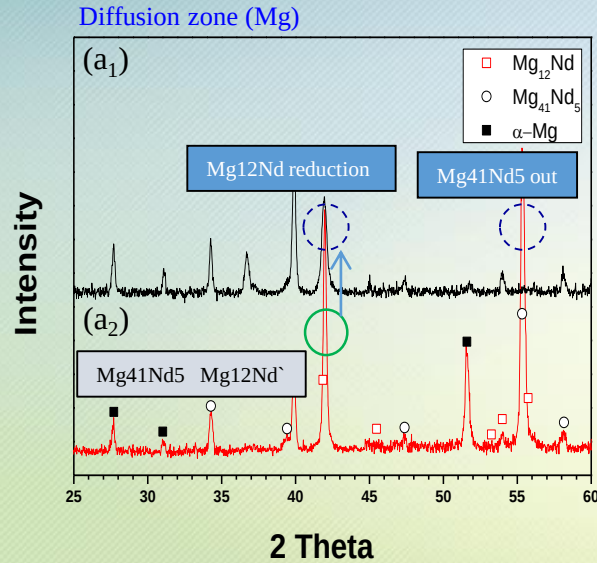
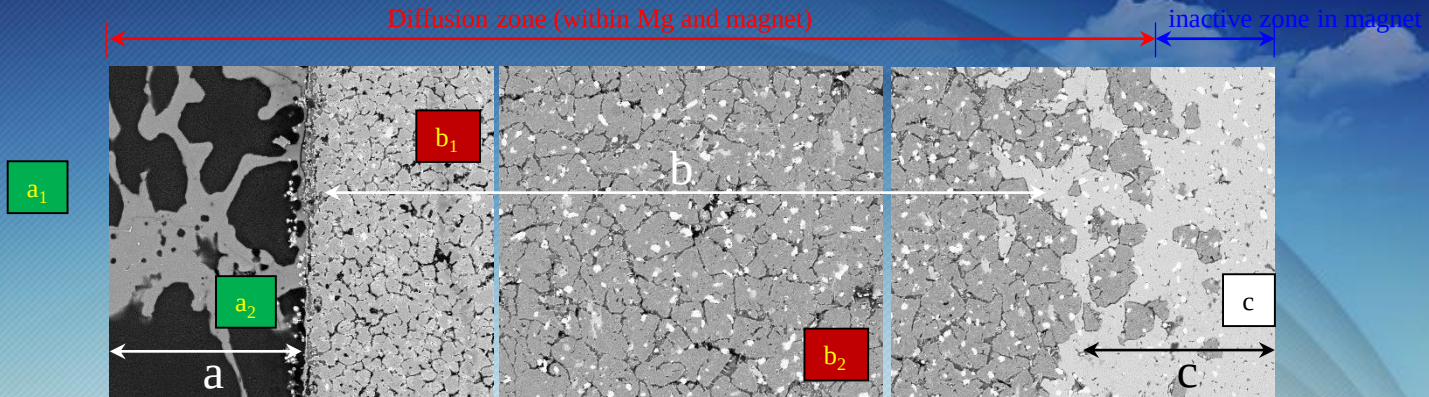


Diffusion layer thickness increment with **the time** from 2 min(a) to 120 min(l)
the temperature from 720°C to 800°C

VOLUME CHANGE of Mg-Nd PHASES with TIME at 720°C

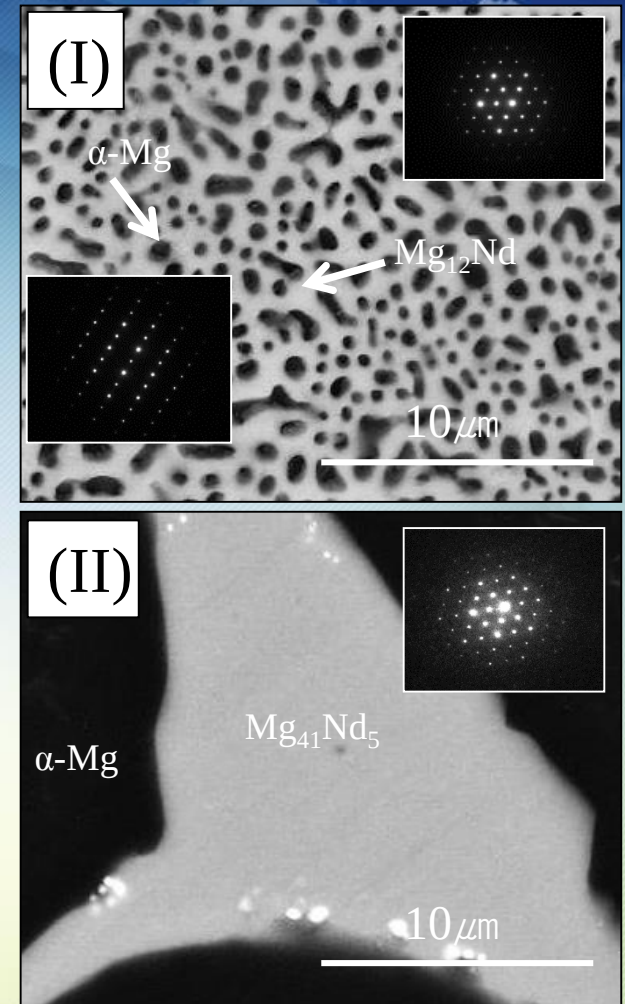
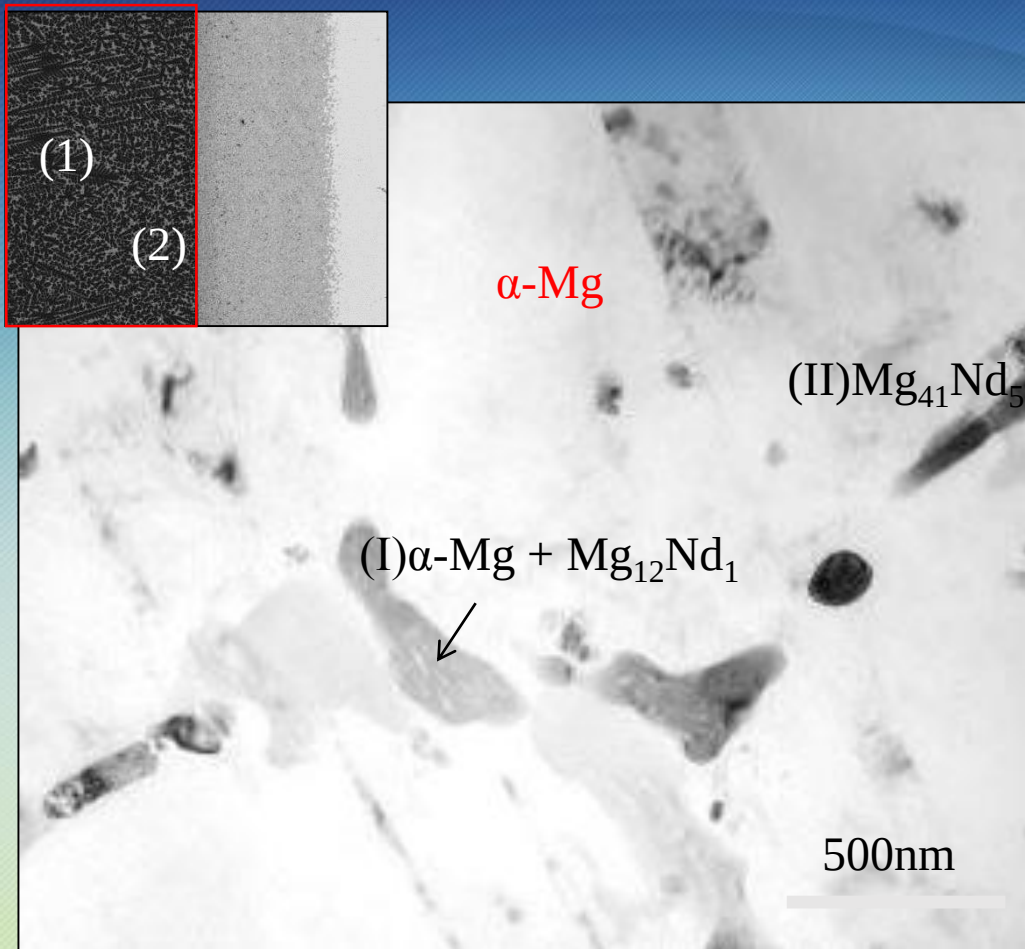


SUMMARY of REACTION



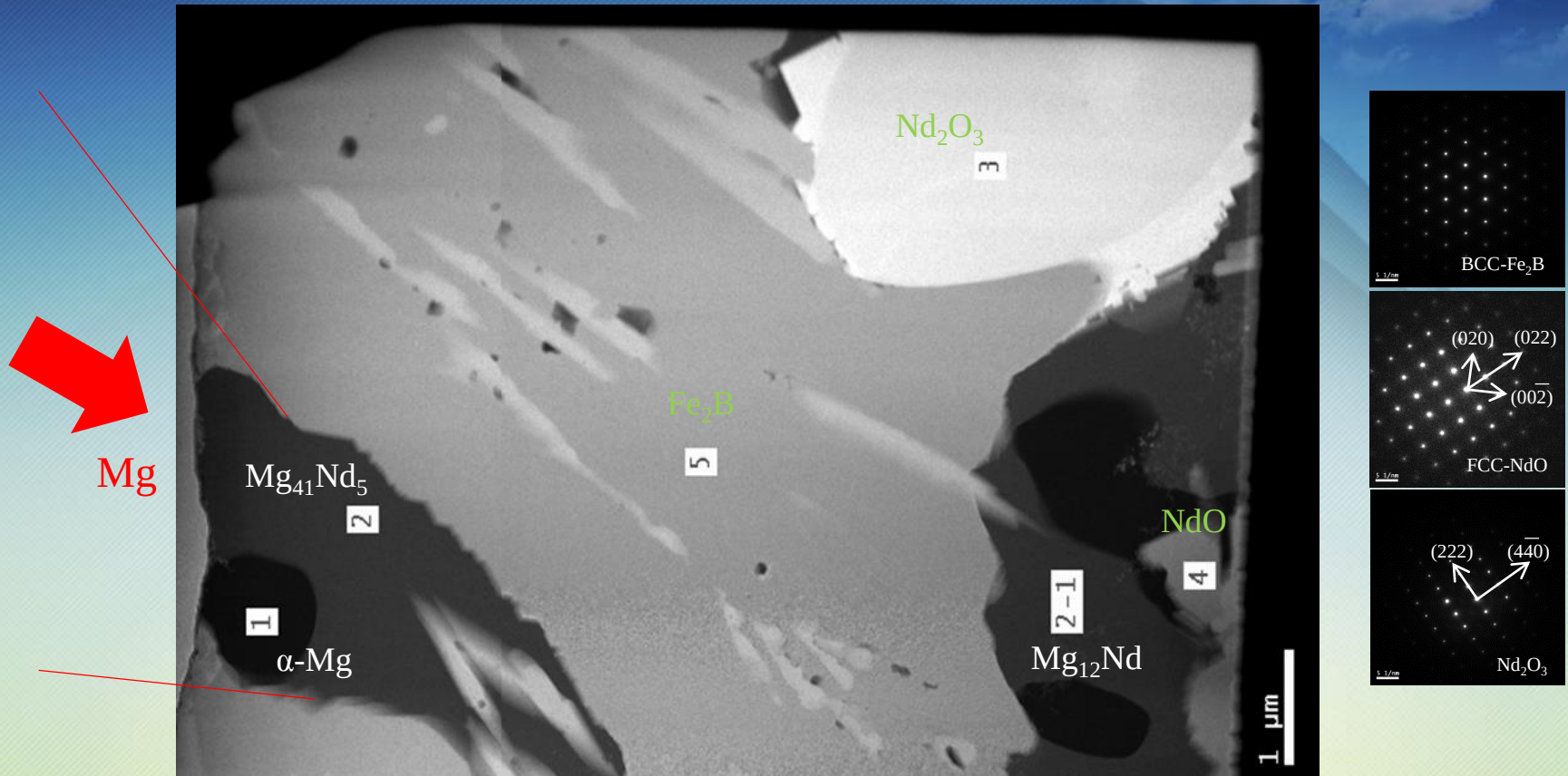
Micro XRD traces showing the various regions from unaffected NdFeB scrap to Mg

TEM MICROGRAPH of Mg REGION after REACTION



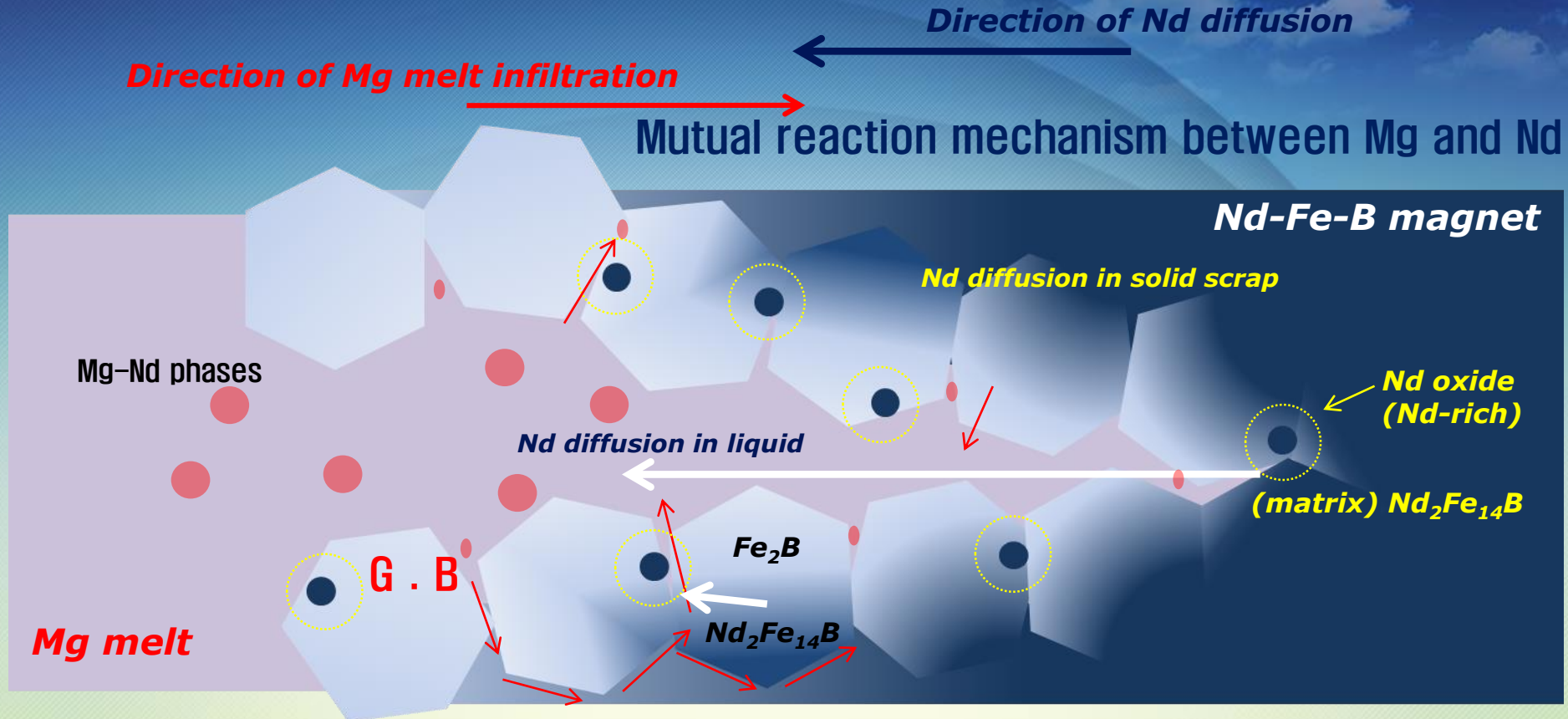
Formation of Mg- $Mg_{12}Nd$ eutectics which is rather fine and spherical than $Mg_{41}Nd_5$

TEM MICROSTRUCTURE within SCRAP



Mg + $\text{Mg}_{41}\text{Nd}_5$ phases form as the Mg passes as a result of diffusion of Nd from the Nd-Fe-B phases, then Fe_2B and Nd_2O_3 remain after the reaction

REACTION MECHANISM



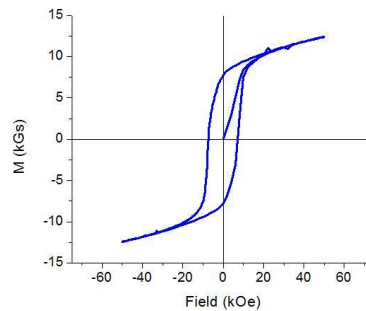
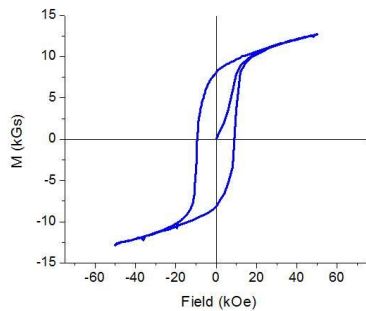
1. As the Nd-Fe-B scrap meets liquid Mg, Nd within Nd rich zone located in the grain boundaries.
2. Mg infiltrates through the G.B, and extracts the Nd from the $\text{Nd}_2\text{Fe}_{14}\text{B}$.
3. With time and temperature, the reaction keep going.

RECOVERY of REE from MAGNET SCRAP and THE MATERIALIZATION RESULTS

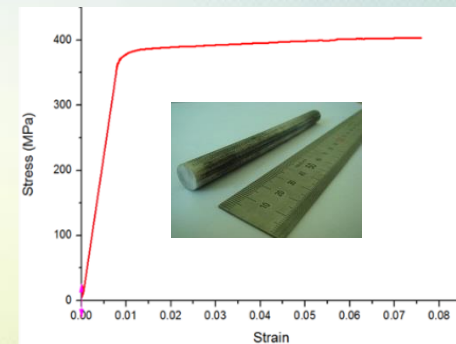
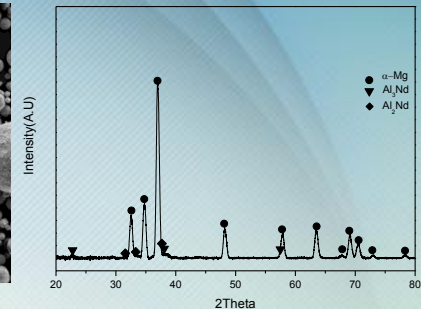
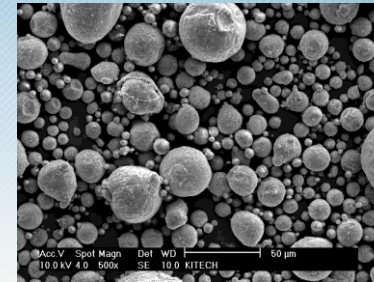
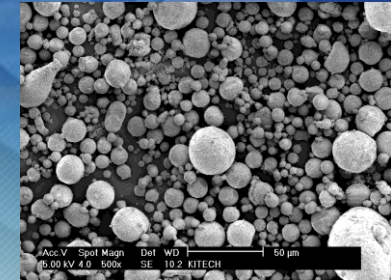
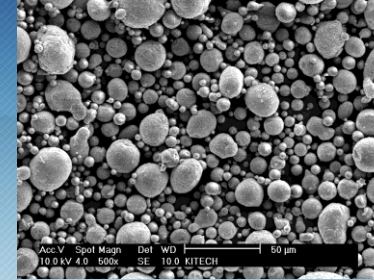
Element	200g scrap	1000g scrap	2000g scrap
Mg	64.6%	50.8%	73.6%
Nd	27.08 %	36.42%	19.28 %
Pr	7.50 %	11.74 %	6.21 %
Dy	0.12 %	0.23%	0.071 %
Fe	0.22 %	0.46 %	0.26 %
Ni	0.34 %	0.081%	0.045 %
B	< 0.001%	< 0.001%	< 0.001%
Total RE Metals	34.70%	48.39%	25.56%

- The scaling up of the process appears to show promising results for recovering high-purity RE metals.

Comparison to magnetic properties of Nd-Fe-B magnet by Recycled RE metal and High pure RE metal



Hysteresis Curves for magnet alloys synthesized from high-purity RE metals refined from oxides (left) RE metals recovered from magnetic scrap (right)



Tensile properties of Mg-5Al-5Zn-2Nd alloy

- UTS of about 400 MPa using Mg-Al-Zn-Nd composition gives a strong assurance on the process, which would be used for recycled Mg-REE to Mg industry .

SUMMARY

AMAES for Environmental equity

- **Intergenerational equity**

Depleting natural resources and degrading ecological health in order to meet short-term objectives can endanger the ability of future generations to meet their needs.

- **Inter-societal equity**

Imbalances of resource use exist between countries and between social and economic groups in a country.

Emergence

Swarm intelligence

*International
collaboration*



**Effective pathway
to solve rare metals paradox**



Thank you
for Your attention!